

SINGLE-LAYER “DOMINO” DIODES VIA OPTOFLUIDIC LITHOGRAPHY FOR ULTRA-LOW REYNOLDS NUMBER APPLICATIONS

Ryan D. Sochol*, Casey C. Glick, Kye Y. Lee, Thomas Brubaker, Albert Lu, Melissa Wah, Shan Gao, Erica Hicks, Ki Tae Wolf, Kosuke Iwai, Luke P. Lee and Liwei Lin
Berkeley Sensor and Actuator Center, University of California, Berkeley, USA

ABSTRACT

Autonomous fluidic components are critical to the advancement of integrated micro/nanofluidic circuitry for lab-on-a-chip applications, such as point-of-care (POC) molecular diagnostics and on-site chemical detection. Previously, a wide range of self-regulating microfluidic components, such as fluidic diodes, have been developed; however, achieving effective functionality at ultra-low Reynolds number (e.g., $Re < 0.05$) has remained a significant challenge. To overcome this issue, here we introduce single-layer microfluidic “domino” diodes, which utilize free-standing rotational microstructures – constructed *in situ* via optofluidic lithography – in order to passively regulate the fluidic resistance based on the flow polarity, thereby enabling flow rectification under ultra-low Re conditions. COMSOL simulation results revealed a theoretical Diodicity (Di) of 31 for a singular domino diode component. Experimental results (for systems with four microstructures) revealed Di 's ranging from 13.0 ± 1.9 to 25.4 ± 1.9 corresponding to $0.025 < Re < 0.030$ and $0.010 < Re < 0.015$ flow, respectively, which represent the largest Di 's reported for $Re < 0.05$ fluid flow.

INTRODUCTION

Next-Generation Microfluidic Circuitry

Autonomous electronic components, such as diodes and transistors, revolutionized the field of electronics. Similarly, the advent of self-regulating microfluidic circuit components could significantly impact both chemical and biological fields [1]. Microfluidic platforms offer a variety of benefits for biochemical applications, such as low reagent volumes, rapid reaction times, and the ability to passively mix, transport, and/or array suspended microbeads and cells [2-4]. Challenges remain, however, for transitioning from the majority of current microfluidic systems that require significant external (i.e., “off-chip”) observation and/or regulation during device operation to fluidic chips that are capable of autonomous (i.e., “on-chip”) self-regulation. The introduction of an externally-controlled, multi-layer elastomeric fluidic valve – often referred to as the *Quake valve* – provided the foundation for the first generation of integrated microfluidic

circuits [5-9]. Such systems, however, require expensive and bulky pressure control equipment, prompting research into self-regulating microfluidic devices which do not require active external control [10-12]. One caveat, however, is that current microfluidic circuit components are primarily constructed by means of multi-layer fabrication processes, which suffer from increased costs, time, and labor versus single-layer microfabrication processes [5-12]. Consequently, research has shifted toward the development of single-layer microfluidic circuit components and systems. For example, recent works have demonstrated microfluidic diodes that utilize suspended microbeads to rectify fluid flow under the low pressures and Re flows required for emerging micro/nanofluidic applications [13-15]. Unfortunately, due to issues stemming from microbead loading and handling within such systems, alternative single-layer methods are needed for microfluidic component construction and operation.

Optofluidic Lithography for Microfluidic Circuitry

Optofluidic lithography processes hold significant potential for advancing micro/nanofluidic circuitry by enabling the fabrication of free-moving elements within single-layer fluidic systems. Previously, researchers have employed optofluidic lithography methods to fabricate free-moving microscale pistons and check valves in two-layer microfluidic systems; however, the efficacy of such components remains limited due to the high pressures that are required for operation [16-19]. Although several single-layer microfluidic circuit components have been successfully fabricated using optofluidic lithography, device functionality required changes in fluidic properties including pH and temperature, which lack the universal applicability associated with regulation *via* alternative flow characteristics, such as flow polarity or pressure [20-22]. To benefit from the advantages of single-layer microfabrication processes, flow polarity-based “on-chip” regulation, and ultra-low Re functionality, here we introduce a new class of single-layer microfluidic diodes that utilize optofluidic lithography to create freely-rotating microstructures *in situ*, which are capable of flow rectification under ultra-low Re conditions (Fig. 1).



Figure 1. Conceptual illustration of the single-layer microfluidic “domino” diode. The diode includes freely-rotating microstructures in series, which pivot about a micropost to passively regulate fluid flow based on the flow polarity. (Left) For forward flow, all of the rotating elements ‘open’ to enable fluid flow through the microchannel. (Right) For reverse flow, the microstructures rotate back to the ‘closed’ position, thereby obstructing fluid flow through the channel.

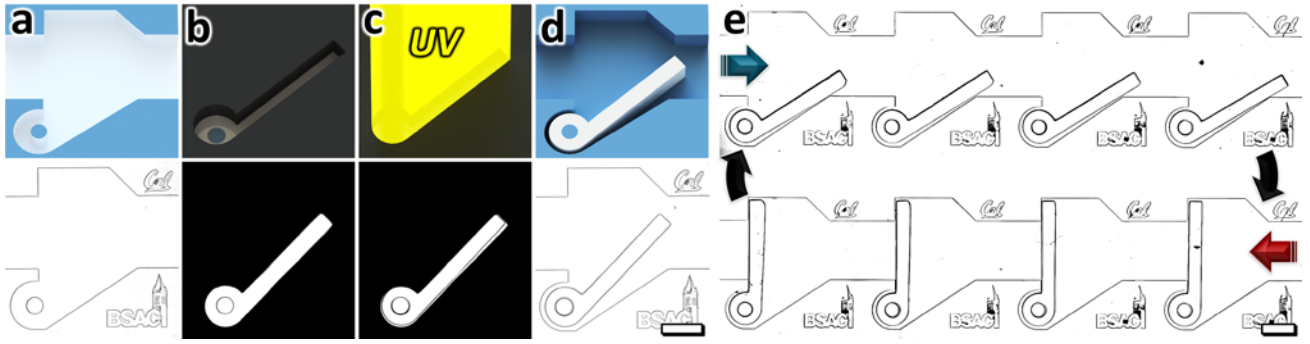


Figure 2. The microfluidic domino diode fabrication process. (a-d) Conceptual illustrations (Top) and experimental results (Bottom) of using optofluidic lithography processes to fabricate single-layer domino diodes. (a) Initially, a photocurable liquid – PEGDA with 1% photoinitiator in this work – is loaded into the microchannel. (b) A photomask is placed in contact with the device. (c) After alignment, the device is exposed to UV for in situ photopolymerization of as many elements in series as desired. Due to an oxygen layer from the surrounding PDMS, the cured microstructures do not bond to the PDMS. (d) Afterward, remaining uncured polymer is evacuated from the device. (e) Brightfield micrographs of fabricated domino diodes under forward (Top) and reverse (Bottom) flow. Scale Bars = 500 μm .

CONCEPT

Figure 1 shows illustrations of the single-layer microfluidic “domino” diode concept. Because the contribution of the non-linear inertial term of the *Navier-Stokes equations* becomes negligible at ultra-low Re [23], here we use freely-rotating microstructures as dynamic resistive elements. Similar to a macroscale “swing check” valve, the microstructures passively pivot about a micropost (100 μm in radius and height) to promote or limit fluid flow based exclusively on the flow polarity. For forward flow, the microstructures rotate clockwise from the *closed* position to the *open* position due to hydrodynamic forces. During this process, the rotation of the serially-arrayed elements resembles dominoes falling in a line, with the key exception that the microstructures do not impact each other. When the flow polarity is reversed, the microstructures rotate counterclockwise to return to the closed position, thereby obstructing fluid flow through the microchannel. This process can be repeated continuously by switching the flow polarity as desired.

FABRICATION

The overall microfluidic system was fabricated using standard soft lithography processes as described previously [2-4]. Micromolded poly(dimethylsiloxane) (PDMS), with microchannel heights of 100 μm , was thermally bonded to glass slides coated with an 80 μm layer of PDMS (*i.e.*, for PDMS-PDMS bonding). Figure 2a-d show sequential conceptual illustrations (*top*) and experimental results (*bottom*) of the fabrication of a freely-rotating microstructure via optofluidic lithography

processes [6, 7]. First, poly(ethylene glycol) diacrylate (PEGDA) with 1% photoinitiator was loaded into the device (Fig. 2a). A photomask was placed in contact with the device and aligned (Fig. 2b). The device was then exposed to UV, resulting in photopolymerization of the microstructures *in situ* (Fig. 2c). Due to the oxygen layer of the surrounding PDMS, the cured PEGDA did not bond to the PDMS, and was slightly smaller in height compared to the microchannel (resulting in a height difference of approximately 5-10 μm). Afterward, the remaining uncured PEGDA was evacuated from the device and replaced with a solution of PEGDA without photoinitiator. Figure 2e shows the domino diode architecture used in this work, which included four elements in series, for both the open (*top*) and closed (*bottom*) positions.

RESULTS AND DISCUSSION

COMSOL Multiphysics Fluid Dynamics Simulations

Three-dimensional fluid pressure field simulations were performed using the commercial finite element analysis software, COMSOL Multiphysics, to characterize a single microfluidic domino diode component (Fig. 3). A sample mesh of the system in the closed state is shown in Figure 3a. For the forward flow (*i.e.*, open state) case, the pressure drop across the system was approximately 2.9 kPa (Fig. 3b). For the reverse flow (*i.e.*, closed state) case, the pressure drop across the system was 90 kPa (Fig. 3c). Thus, the pressure field simulation results revealed a theoretical Di (*i.e.*, the ratio of the reverse flow pressure drop over the forward flow pressure drop) of approximately 31 for a singular domino diode component.

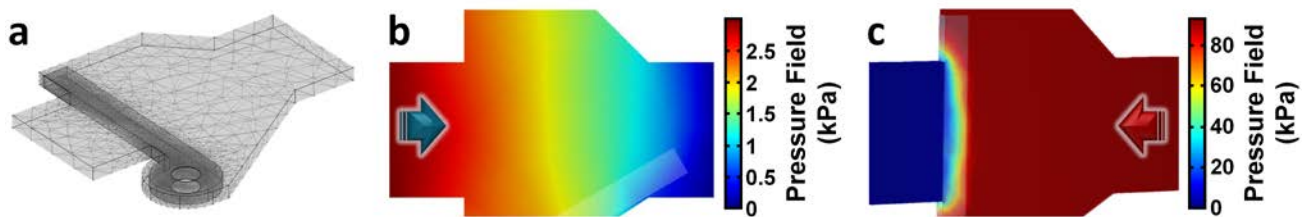


Figure 3. Three-dimensional COMSOL Multiphysics simulations for a single microfluidic domino diode component. (a) Sample mesh of the modeled diodic component. (b) Pressure field simulation results for the forward flow (*i.e.*, open state) case. (c) Pressure field simulation results for the reverse flow (*i.e.*, closed state) case.

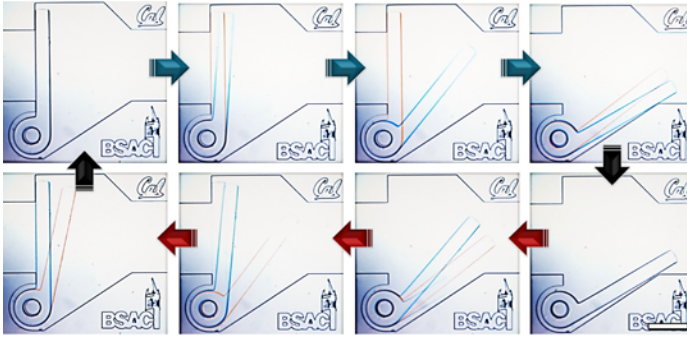


Figure 4. Sequential brightfield micrographs captured of experimental results for microfluidic domino diode dynamics. (Top) Forward flow case. (Bottom) Reverse flow case. This process occurs autonomously via external switches in the flow polarity. The process can be repeated continuously as desired. Orange/blue microstructures denote two distinct time-points within one second. Scale Bar = 500 μm .

Experimental Results

Figure 4 shows experimental results for the rotational dynamics of a single domino diode microstructure in response to changes in the flow polarity. Under forward flow, the microstructure rotated about the micropost to switch from the closed state to the open state (Fig. 4 – top). After the flow polarity was reversed, the microstructure rotated in the opposite direction to restore the closed state (Fig. 4 – bottom). Microstructure rotation was observed at the lowest Re achievable with our setup (i.e., $Re \approx 0.01$). Multiple device runs revealed this process to be repeatable.

Quantified Di results for single-layer microfluidic domino diodes with four microstructures in series are presented in Figure 5. Experiments were performed for Re varying from 0.01 to 0.03. For Re ranging from 0.01 to 0.015, experimental results revealed an average Di of 25.4 ± 1.9 . The Di performance was found to decrease with increasing Re . For example, the Di decreased significantly to 13.0 ± 0.4 ($p < 10^{-4}$) for Re varying from 0.025 to 0.03. This trend is in contrast to the Di behavior of our prior microbead-based diodes [14, 15], which revealed higher Di with increasing Re . One possible basis for this phenomenon is the potential deformation of the PDMS at larger pressures/ Re , which would increase the micro-channel height. As a result, leakage of fluid past microstructures in the closed position (i.e., during reverse flows) would be promoted with increasing Re , thereby reducing the resistive effects of the blocking microstructures. For the current study, the devices included PDMS mixed at a 10:1 (base : curing agent) ratio and were cured at 55 $^{\circ}\text{C}$; however, we expect that increasing the Young's Modulus of the PDMS (e.g., by using PDMS that is mixed with a lower base : curing agent ratio and/or curing the PDMS at higher temperatures) would limit the potential for deformation at higher pressures/ Re .

For all of the Re flows tested, the experimental results revealed Di 's that were lower than the theoretical Di . Specifically, although the simulation results yielded a theoretical Di of approximately 31, the largest experimental Di observed was 25.4 ± 1.9 (i.e., for $0.010 < Re < 0.015$). Several differences between the COMSOL model and the fabricated devices could have contributed to the reduction in Di performance observed during

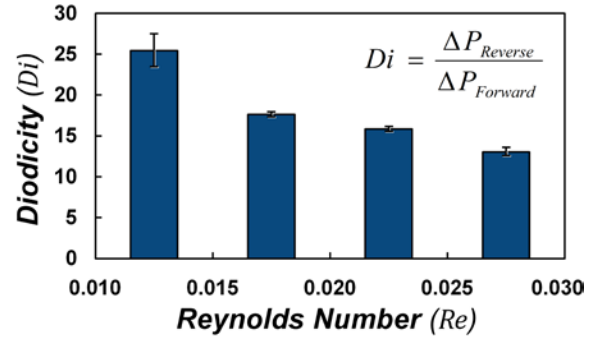


Figure 5. Quantified experimental results for Diodicity (Di) versus Reynolds number (Re) corresponding to a microfluidic domino diode with four photoexposed rotational microstructures. $\Delta P_{Forward}$ and $\Delta P_{Reverse}$ denote the pressure drop for the forward and reverse flow cases, respectively. Error Bars denote s.e.m.

experimental testing. As discussed previously, it is possible that the PDMS of the diode may have deformed during operation, leading to increased leakage past closed microstructures. Additionally, photopolymerized PEGDA can have a porous structure, which could permit fluid flow through closed microstructures. Because the simulations modeled rigid system components, the aforementioned phenomena would not be accounted for in the theoretical Di prediction. These results suggest that adaptations of the current methodology that utilize stiffer and/or non-porous materials could enhance the Di performance.

CONCLUSIONS

Integrated micro/nanofluidic circuitry offers the potential to revolutionize POC lab-on-a-chip applications, rendering robust fluidic components that are capable of autonomous on-chip functionalities (at ultra-low Re) a critical priority. In this work, we presented a single-layer microfluidic domino diode that utilized freely-rotating microstructures – fabricated *in situ* via optofluidic lithography – to rectify fluid flow under $Re < 0.03$ conditions. Experimental testing revealed Di 's within the range of 13.0 ± 1.9 (i.e., for $0.025 < Re < 0.030$) to 25.4 ± 1.9 (i.e., for $0.010 < Re < 0.015$). These results mark an improvement over current microfluidic diodes that exhibit limited Di capabilities for $Re < 0.03$ fluid flow [13-15]. Several adaptations of the presented technique could improve the efficacy of the diode. For example, although the microfluidic domino diode used in the current study included four rotating microstructures that were arrayed in series, the same methodology could be employed to simultaneously fabricate higher numbers of microstructures in series to significantly increase the pressure drop associated with the reverse flow case. Additionally, the use of PDMS with higher Young's Moduli and/or photocurable polymers with lower porosities could also improve the Di performance. Here, optofluidic lithography processes were employed to resolve a microfluidic circuit component that is capable of flow rectification; however, the presented methodology could be adapted to achieve a new class of single-layer microfluidic circuit elements with diverse functionalities for a variety of chemical and biological applications.

ACKNOWLEDGEMENTS

The authors greatly appreciate the help and support of Nazly Pirmoradi, Jonathan Lei, Vishnu Jayaprakash, and Paul Lum. The authors also thank the members of the Liwei Lin Laboratory, the Luke P. Lee *Biologically-inspired Photonics-Optofluidics-Electronics Technology and Science (BioPOETS)* Laboratory, the Shoji Takeuchi *Biohybrid System* Laboratory, and the *Micro Mechanical Methods for Biology (M³B)* Laboratory Program. This research is supported in part by the DARPA N/MEMS program under the Micro/Nano Fluidics Fundamentals Focus (MF3) center.

REFERENCES

- [1] B. Mosadegh, T. Bersano-Begey, J. Y. Park, M. A. Burns, and S. Takayama, "Next-generation integrated microfluidic circuits," *Lab on a Chip*, vol. 11, pp. 2813-2818, 2011.
- [2] R. D. Sochol, B. P. Casavant, M. E. Dueck, L. P. Lee, and L. Lin, "A dynamic bead-based microarray for parallel DNA detection," *Journal of Micromechanics and Microengineering*, vol. 21, p. 054019, 2011.
- [3] R. D. Sochol, S. Li, L. P. Lee, and L. Lin, "Continuous flow multi-stage microfluidic reactors via hydrodynamic microparticle railing," *Lab on a Chip*, vol. 12, pp. 4168-4177, 2012.
- [4] R. D. Sochol, M. E. Dueck, S. Li, L. P. Lee, and L. Lin, "Hydrodynamic resettability for a microparticle arraying system," *Lab on a Chip*, vol. 12, pp. 5051-5056, 2012.
- [5] M. A. Unger, H. P. Chou, T. Thorsen, A. Scherer, and S. R. Quake, "Monolithic microfabricated valves and pumps by multilayer soft lithography," *Science*, vol. 288, pp. 113-116, 2000.
- [6] T. Thorsen, S. J. Maerkl, and S. R. Quake, "Microfluidic large-scale integration," *Science*, vol. 298, pp. 580-584, 2002.
- [7] J. W. Hong, V. Studer, G. Hang, W. F. Anderson, and S. R. Quake, "A nanoliter-scale nucleic acid processor with parallel architecture," *Nature Biotechnology*, vol. 22, pp. 435-439, 2004.
- [8] C.-C. Lee, G. Sui, A. Elizarov, C. J. Shu, Y.-S. Shin, A. N. Dooley, J. Huang, A. Daridon, P. Wyatt, D. Stout, H. C. Kolb, O. N. Witte, N. Satyamurthy, J. R. Heath, M. E. Phelps, S. R. Quake, and H.-R. Tseng, "Multistep synthesis of a radiolabeled imaging probe using integrated microfluidics," *Science*, vol. 310, pp. 1793-1796, 2005.
- [9] J. Wang, H. C. Fan, B. Behr, and S. R. Quake, "Genome-wide single-cell analysis of recombination activity and de novo mutation rates in human sperm," *Cell*, vol. 150, pp. 402-412, 2012.
- [10] D. C. Leslie, C. J. Easley, E. Seker, J. M. Karlinsey, M. Utz, M. R. Begley, and J. P. Landers, "Frequency-specific flow control in microfluidic circuits with passive elastomeric features," *Nature Physics*, vol. 5, pp. 231-235, 2009.
- [11] J. A. Weaver, J. Melin, D. Stark, S. R. Quake, and M. A. Horowitz, "Static control logic for microfluidic devices using pressure-gain valves," *Nature Physics*, vol. 6, pp. 218-223, 2010.
- [12] B. Mosadegh, C.-H. Kuo, Y.-C. Tung, Y.-S. Torisawa, T. Bersano-Begey, H. Tavana, and S. Takayama, "Integrated elastomeric components for autonomous regulation of sequential and oscillatory flow switching in microfluidic devices," *Nature Physics*, vol. 6, pp. 433-437, 2010.
- [13] K. Ou, J. Jackson, H. Burt, and M. Chiao, "Microspheres as resistive elements in a check valve for low pressure and low flow rate conditions," *Lab on a Chip*, 2012.
- [14] R. D. Sochol, K. Iwai, J. Lei, D. Lingam, L. P. Lee, and L. Lin, "A single-microbead-based microfluidic diode for ultra-low Reynolds Number applications," in *Proceedings of the IEEE 25th International Conference on Micro Electro Mechanical Systems (MEMS 2012)*, Paris, France, 2012.
- [15] R. D. Sochol, J. Lei, A. Lu, E. L. Hicks, S. Gao, V. Menon, L. P. Lee, and L. Lin, "Circular microchannels enhance diodicity performance at ultra-low reynolds number for microfluidic bead-based diodes," in *Proceedings of MicroTAS 2012*, Okinawa, Japan, 2012.
- [16] E. F. Hasselbrink, T. J. Shepodd, and J. E. Rehm, "High-pressure microfluidic control in lab-on-a-chip devices using mobile polymer monoliths," *Analytical Chemistry*, vol. 74, pp. 4913-4918, 2002.
- [17] B. J. Kirby, T. J. Shepodd, and E. F. Hasselbrink Jr, "Voltage-addressable on/off microvalves for high-pressure microchip separations," *Journal of Chromatography A*, vol. 979, pp. 147-154, 2002.
- [18] D. S. Reichmuth, T. J. Shepodd, and B. J. Kirby, "On-Chip high-pressure picoliter injector for pressure-driven flow through porous media," *Analytical Chemistry*, vol. 76, pp. 5063-5068, 2004.
- [19] B. J. Kirby, D. S. Reichmuth, R. F. Renzi, T. J. Shepodd, and B. J. Wiedenman, "Microfluidic routing of aqueous and organic flows at high pressures: fabrication and characterization of integrated polymer microvalve elements," *Lab on a Chip*, vol. 5, pp. 184-190, 2005.
- [20] D. J. Beebe, J. S. Moore, J. M. Bauer, Q. Yu, R. H. Liu, C. Devadoss, and B. H. Jo, "Functional hydrogel structures for autonomous flow control inside microfluidic channels," *Nature*, vol. 404, pp. 588-590, 2000.
- [21] M. Vazquez and B. Paull, "Review on recent and advanced applications of monoliths and related porous polymer gels in micro-fluidic devices," *Analytica Chimica Acta*, vol. 668, pp. 100-113, 2010.
- [22] D. Kim and D. J. Beebe, "A bi-polymer micro one-way valve," *Sensors and Actuators: A-Physical*, vol. 136, pp. 426-433, 2007.
- [23] J. Loverich, I. Kanno, and H. Kotera, "Single-step replicable microfluidic check valve for rectifying and sensing low Reynolds number flow," *Microfluidics and Nanofluidics*, vol. 3, pp. 427-435, 2007.

CONTACT

*R. D. Sochol, tel: +1-410-935-8971; rsochol@gmail.com